

RIANA MALHOTRA

647-594-1845 | riana.malhotra@mail.utoronto.ca | [LinkedIn](#) | [Github](#)

EDUCATION

Bachelor of Applied Science - Computer Engineering + PEY Co-op
University of Toronto

Expected May 2029

- UofT Engineering International Scholar Award Recipient
- Minors: Artificial Intelligence and Engineering Business

TECHNICAL SKILLS

Programming Languages: C, C++, C#, Python, Java, Verilog, NIOS-V Assembly, MATLAB, JavaScript, SQL

Embedded & Firmware: STM32, QNX RTOS, FPGA Design, PCB Design

Software & Testing: Git, Linux/UNIX, REST APIs

Hardware Design: Altium Designer, LTSpice, ANSYS, CAD

EXPERIENCE

Production Test Engineer Intern | Trimble Applanix

May 2025 - Aug 2025, May 2026 - Aug 2026

- Programmed, validated, and debugged 50+ QNX-based firmware builds for airborne and marine GNSS/INS platforms, resolving boot, SD/NAND, and hardware-software integration issues prior to production.
- Developed 15+ C/C# utilities to automate firmware flashing and inertial module provisioning for GNSS/INS devices, reducing production programming time by ~20% through workflow automation.
- Executed regression testing and system validation for GNSS/INS platforms using CAN, UART, oscilloscopes, and logic analyzers, identifying firmware and communication issues while partnering with QA to improve calibration accuracy and system reliability.

President | University of Toronto Aviation Club

May 2026 - Present

- Founded, helped establish and shape the UofT Aviation Club by defining its mission, structure, and long-term vision to promote aviation and aerospace engagement on campus.
- Lead club operations, events, and technical initiatives, including workshops, speaker sessions, and outreach activities aimed at increasing student participation in aviation-related fields.
- Coordinate executive teams, manage cross-functional planning, and build external partnerships with industry and academic stakeholders to expand opportunities and resources for UofT students.

Software Member - Avionics Rocketry Subteam | University of Toronto Aerospace Team

Sep 2023 - Present

- Developed embedded C software for STM32 microcontrollers, implementing real-time control logic for sensor and actuator subsystems to improve system responsiveness and stability.
- Built Python-based integration and communication test tools to validate embedded data paths, identifying protocol and timing issues before hardware integration.
- Debugged STM32 peripherals (GPIO, timers, UART, and serial interfaces) to resolve hardware-software integration issues and ensure reliable sensor data acquisition.
- Contributed to PCB design, schematic review, and board bring-up in Altium, validating embedded interfaces through circuit debugging and hardware testing.

Systems Member - Autonomous Rover Team | University of Toronto Robotics Association

Jan 2024 - Dec 2024

- Developed automated tests in C++ and Python to check rover systems and ensure reliable operation.
- Created Python scripts to help identify bugs and troubleshoot issues during rover testing and simulations.
- Tested the rover's autonomous navigation by working with ROS, Arduino, sensors, and real-time data.
- Collaborated with team members to test rover performance, identify issues, and improve system functionality through iterative testing.

Structural & Fabrication Member | Blue Sky Solar Racing (University of Toronto)

Sep 2024 - Apr 2025

- Designed and analyzed solar-car structural components using CATIA and ANSYS FEA, optimizing strength, load distribution, and durability under dynamic race conditions.
- Fabricated and assembled carbon-fiber composite structures using precision machining and hand tools, ensuring tight tolerances and high strength-to-weight performance.
- Iterated designs through simulation and physical testing feedback loops, improving structural reliability and ensuring compliance with safety and performance requirements.

PROJECTS

TransitFlow - GIS for Commuters | ECE297 - Software Design and Communication Jan 2026 - Apr 2026

- Built a C++ GIS-based software commuter navigation system using A*, Dijkstra, and metaheuristic optimization algorithms for route planning.
- Developed multi-threaded optimization pipeline using Greedy, Ant Colony Optimization, 2-opt, OR-opt, and Simulated Annealing, improving solutions by 8% over baseline.

FPGA DJ Booth - Embedded Audio Processing System | ECE243 - Computer Organization Mar 2026 - Apr 2026

- Developed FPGA-based audio workstation using Nios V processor and assembly, integrating microphone input, VGA display, switches, and peripherals.
- Implemented real-time audio effects including echo, delay, reverb, volume control, and waveform visualization.

BreadCrumb Trail - Handheld GNSS Navigation System | MakeUofT Hardware Hackathon Feb 2026

- Won Best Use of Arduino Uno Q at MakeUofT; developed an embedded GNSS navigation device integrating UART, compass, OLED display, and GSM communication.
- Implemented breadcrumb logging and spherical navigation algorithms with real-time visualization, enabling route tracking and waypoint guidance.

Pong Game - FPGA Real-Time VGA Graphics | ECE241 - Digital Systems Sep 2025 - Nov 2025

- Implemented a Verilog-based Pong game on DE1-SoC FPGA featuring FSM-driven game logic, collision handling, position tracking, and VGA output generation.
- Integrated hardware modules for keyboard input, clock-driven motion control, and real-time score display using synchronous digital design principles.

ClawBot - Autonomous Flag-Seeking Robot | UTRA Hacks Feb 2025

- Built an autonomous Arduino robot integrating color sensors, ultrasonic sensors, PID control, Bluetooth communication, and a servo-actuated claw mechanism.
- Optimized real-time navigation and obstacle avoidance through sensor fusion and embedded control algorithms.
- Integrated autonomous path following, obstacle avoidance, and servo-based flag capture into a unified control system, debugging and tuning the robot under a 24-hour development constraint.

Morse Code Blink Decoder - FPGA & Computer Vision Nov 2024 - Dec 2024

- Developed a real-time blink-to-Morse communication system using OpenCV and Python, classifying eye blinks as dots and dashes for hands-free text input.
- Designed and implemented a Verilog finite state machine (FSM) on an FPGA to parse Morse sequences and decode them into corresponding characters.
- Integrated computer vision with FPGA hardware through UART-to-USB serial communication and implemented VGA output to display decoded text in real time.

TelMe - GenAI Telecom Support Platform | NewHacks Oct 2024

- Developed an AI-powered telecom support platform in a 24-hour hackathon, combining a conversational chatbot with personalized customer insights to improve onboarding and retention.
- Built Python/Flask APIs powering real-time chatbot interactions and customer-facing ML features, integrating backend services with a web-based frontend.
- Implemented K-means clustering to segment customers and generate personalized telecom plan recommendations based on usage patterns and customer needs.
- Developed an interactive customer portal delivering data-driven recommendations and insights, demonstrating end-to-end development across machine learning, APIs, and web applications.

Accessible Stationery Tools | APS112 - Engineering Strategies & Practice II Jan 2024 - Apr 2024

- Designed a one-handed stationery system for users with mobility constraints, using stakeholder requirements, brainstorming, SCAMPER, morph-chart analysis, and Pugh evaluation.
- Developed a Rubber Base Glide concept incorporating a stabilizing rubber base and sliding mechanism to improve precision during one-handed use.
- Created a functional SOLIDWORKS prototype and developed a testing plan evaluating accuracy, portability and usability.

ADDITIONAL INFORMATION

- **Aviation:** Private Pilot License Candidate | 25+ flight hours on Cessna 172S | ROC-A Radio Licence
- **Music:** Grade 8 Classical Acoustic Guitarist, formally trained through the London College of Music; Award of Excellence recipient with high distinctions in guitar examinations.

Outlier · Freelance

Jan 2025 - May 2026 · 1 yr 5 mos

San Francisco, California, United States · Remote

- Trained and evaluated large language models (LLMs) on software engineering tasks including algorithmic problem solving, code synthesis, debugging, and reasoning over data structures
- Performed systematic error analysis on model outputs to identify logical inconsistencies, and edge-case failures; provided targeted feedback to improve model alignment and reliability
- Designed structured prompts, unit-style test cases, and evaluation benchmarks to measure functional correctness, logical soundness, and robustness of model responses
- Reviewed and validated AI-generated code for syntactic correctness, algorithmic efficiency, adherence to language standards, and best practices across Python, C/C++, and Java

GRASIM INDUSTRIES LIMITED (ADITYA BIRLA GROUP) logo
Web Developer

GRASIM INDUSTRIES LIMITED (ADITYA BIRLA GROUP) · Internship

Jan 2023 - Aug 2023 · 8 mos

Mumbai, Maharashtra, India · On-site

- Created and maintained dummy websites and simple web applications using HTML, CSS, and JavaScript, helping the team prototype features and test new designs.
- Improved user experience by updating UI elements like navigation menus, color schemes, and responsive layouts using Figma, making sites easier to use on different devices.
- Assisted in deploying websites to the cloud using AWS, gaining hands-on experience with cloud hosting and improving site uptime and performance.
- Collaborated with the development team to troubleshoot bugs and optimize page load speeds, contributing to smoother user interactions.
- Learned and applied version control basics to manage code changes and work effectively within the team.
- Participated in weekly team meetings, sharing progress updates and receiving feedback to improve technical skills and project outcomes.

Cascading Style Sheets (CSS), Figma (Software) and +5 skills

MySkin Laser Clinics logo
Data Analyst

MySkin Laser Clinics · Internship

Jan 2022 - Apr 2022 · 4 mos

Mumbai, Maharashtra, India · Hybrid

- Collected and organized data from diagnostic tools, creating a reliable database to support clinical decision-making.
- Used Excel and SQL to analyze client data, identifying patterns that assisted in correlating symptoms with diagnoses.
- Maintained accuracy and integrity of image and numerical data sets, ensuring clinicians had easy access to up-to-date information.
- Collaborated with the clinical team to verify data consistency and resolve discrepancies, improving overall data quality.

SQL and Microsoft Excel

Science Box Ltd logo
Mentor

Science Box Ltd · Internship

Sep 2021 - Dec 2021 · 4 mos

Hamilton, Waikato, New Zealand · Hybrid

- Led workshops on software tools and coding, demonstrating how simple programming concepts can lead to innovative scientific projects.
- Conducted CAD workshops to teach participants how to design and prototype scientific models, fostering creativity and technical skills.
- Collaborated with students from Australia, New Zealand, India, Bangladesh, and Kenya to promote scientific ideation, exploration, and discovery through cross-cultural projects.
- Developed engaging content that combined hands-on experimentation with digital tools, enhancing participants' understanding of science and technology.

Interpersonal Skills, Computer-Aided Design (CAD) and +3 skills

Biomimetic Robotics & Agri-Tech Research

bBot · Internship

May 2021 - Aug 2021 · 4 mos

Toronto, Ontario, Canada · Remote

- Worked with a Hatchery Startup @ UofT
- UofT- Researched and analyzed biomimetic robotics and machine learning applications for Agri-tech, identifying promising approaches to improve robotic pollination systems.
- Performed mathematical modeling and calculations to optimize vertical farm operations, contributing to increased resource efficiency and crop yield projections.
- Investigated cutting-edge sensors and automation technologies, synthesizing findings to inform design improvements in robotic pollinator prototypes.
- Collaborated remotely with a multidisciplinary team to tackle challenges in precision, navigation, and adaptability, supporting iterative development of autonomous pollination robots.

Hardware Engineering, Robotics and +3 skills